

2.6–2.7 LU, LDU, Permutations, and Transposes

Concise exam revision notes for elimination factorizations and matrix operation rules.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Elimination can be stored as a factorization. Instead of redoing row operations, write the matrix as products of simple matrices.

Without row exchanges, elimination gives

$$A = LU.$$

If the pivots are separated into a diagonal matrix D , then

$$A = LDU.$$

Here L stores elimination multipliers, D stores pivots, and U stores the final upper triangular structure.

2 Solving With LU

To solve

$$Ax = b, \quad A = LU,$$

do two triangular solves:

$$Lc = b, \quad Ux = c.$$

Step	Equation	Meaning
Forward substitution	$Lc = b$	Apply stored elimination.
Back substitution	$Ux = c$	Solve the upper triangular system.

Exam memory. LU is useful because solving triangular systems is faster than starting elimination again.

3 Permutations

If a row exchange is needed, use a permutation matrix:

$$PA = LU.$$

Permutation matrices reorder rows and satisfy

$$P^{-1} = P^T, \quad P^T P = I.$$

Fact	Meaning
$P^{-1} = P^T$	Undoing a permutation is another permutation.
$P^T P = I$	Permutations preserve lengths and dot products.
PA	Reorders rows of A .
AP	Reorders columns of A .

4 Transpose and Inverse Rules

Order reverses when transposing or inverting products:

$$(AB)^T = B^T A^T, \quad (AB)^{-1} = B^{-1} A^{-1}.$$

For an inverse transpose,

$$(A^{-1})^T = (A^T)^{-1}.$$

Common mistake. Do not transpose a product in the same order. The order must reverse.

5 Symmetric Matrices

A matrix is symmetric when

$$A^T = A.$$

The product $A^T A$ is always symmetric:

$$(A^T A)^T = A^T A.$$

The dot-product identity behind this is

$$(Ax)^T y = x^T A^T y.$$

6 Block Transposes

Transpose works block by block and also swaps block positions:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^T = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}.$$

This is the same rule as ordinary transposes: rows become columns.

7 Banded Matrices

Banded matrices have nonzeros near the diagonal, so elimination can be cheaper when the band structure is preserved. A common example is tridiagonal structure:

$$\begin{bmatrix} * & * & 0 & 0 \\ * & * & * & 0 \\ 0 & * & * & * \\ 0 & 0 & * & * \end{bmatrix}.$$

The exam habit is to use the structure instead of treating every entry as dense.

8 Common Mistakes

Mistake	Correct exam habit
Forgetting row exchanges in LU.	Use $PA = LU$ when pivots require swaps.
Solving $LUx = b$ in one jump.	First solve $Lc = b$, then $Ux = c$.
Writing $(AB)^T = A^T B^T$.	Reverse order: $B^T A^T$.
Assuming $A^T A = AA^T$.	Both are symmetric, but usually different.
Ignoring zeros in a banded matrix.	Use the sparsity pattern to simplify work.

9 Final Exam Checklist

Checklist.

1. Check whether row exchanges are needed.
2. Use $A = LU$ if no row exchanges occur.
3. Use $PA = LU$ if row exchanges occur.
4. Solve $Lc = b$ by forward substitution.
5. Solve $Ux = c$ by back substitution.
6. Reverse order when transposing or inverting products.
7. Check symmetry with $A^T = A$.
8. Use banded or triangular structure when present.